



Information Retrieval



Academic Year 2025/2026

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**Master Degree in
Computer Engineering**

Information Retrieval

- Architecture of the modern Information Retrieval Systems
- Data Structures for managing data
- Algorithms for retrieving data
- Boolean Retrieval Systems
- Free Text Retrieval System
- Word Embedding, Word2Vec and Doc2Vec
- Solr and MongoDB ^{reduced}
- Augmentation ^{new}
- Text Classification ^{new}
- Content Image Retrieval Systems

The lectures

- From the 30th of September till the end of December
 - Monday (02.00PM - 04.00PM)
 - Tuesday (04.00PM - 06.00PM)
 - Friday (No Lecture)
- 6 CFU
- 60 hours
 - 38 hours: lectures
 - 22 hours: laboratory

The WebPage

<https://elearning.unipv.it/course/view.php?id=9888>

- Notifications and Alerts
- Teaching Materials



Login with username and password for all university services

The Book (1)

Christopher D. Manning, Prabhakar Raghavan and Hinrich Schütze,
Introduction to Information Retrieval, Cambridge University Press. 2008

<https://nlp.stanford.edu/IR-book/>

Chapters 1, 2, 3, 4, 5, 6, 7, 13, 19, 20

The Book (2)

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein

Introduction to algorithms

The Book (3)

Online resources

The Exam

Written

- 10 open questions
- One hour

Project discussion

The Project

Implementing one or more topics described during lectures

for example: similarities, error detection, vector space model, ...

Some hints are available on the web page

Writing a paper describing the project

One student per project

The Survey

- Proposals, criticisms and suggestions will be welcome
- You will be given instructions on how to answer the questionnaire
- Not mandatory but strongly recommended
- Anonymous

Contacts

For any information and appointments: **luigi.santangelo@unipv.it**